

Intravenous drug use in neonatal intensive care units

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ABSTRACT

Objectives Intravenous drug use in neonates is frequent and prone to medication errors. The aim of this study was to describe the intravenous drugs most frequently used in Spanish Neonatal Intensive Care Units (NICU), their preparation and the implementation rate of standardised concentration infusions.

Methods We conducted an observational multicentre study based on a survey sent by email to nine Spanish NICUs during January and February 2018. We collected data describing the intravenous drugs frequently used in neonates and their preparation. A descriptive analysis of the medicines reported (and their preparation) was performed, to assess how frequently standard concentrations were used and how medications were prepared in central pharmacies.

Results Overall, 69 different drugs were reported by participating NICUs. Of these, 33% (n=23) were not approved for use in neonates and 38% (n=26) corresponded to high-alert medications, according to the Institute for Safe Medication Practices. A mean of only 63.5% of intravenous medicines were standardised. The standard-concentration implementation rate was somewhat higher for intermittent (mean 74.1%) than continuous (mean 42.9%) infusions. Notably, infusions were more commonly prepared on wards than in hospital pharmacies.

Conclusions Intravenous drug use in NICUs has been identified as a high-risk process, and error-reduction strategies (such as concentration standardisation) have been recommended. Further data are necessary to design the most suitable intervention in our country (Spain), but institutional initiatives are needed to achieve this.

INTRODUCTION

Patient safety has been defined as “the reduction of risk of unnecessary harm associated with healthcare to an acceptable minimum”. Healthcare-associated harm is described as “harm arising from or associated with plans or actions taken during the provision of healthcare, rather than an underlying disease or injury”.¹ Medication errors are three times more likely to occur in children than in adults.² Notably, errors with the potential to cause harm are eight times more likely to occur in a neonatal intensive care unit (NICU) than in adult settings in the hospital.³ Neonates are a particularly vulnerable population and are at particular risk of harm from medication errors because of weight-based dosing, off-label drug usage, their rapidly changing body size, their inability to communicate with providers, and their physiological immaturity.^{4–6}

Fortunately, most medication errors do not result in harm to neonatal patients.⁵ In a retrospective analysis of medication errors reported in NICUs,

only 3.6% were harmful, and just one case resulted in death.³

Drugs are frequently used in children in conditions that differ from those authorised in their product specifications, due to the lack of specific data from clinical trials in children. In some instances, the use of unlicensed drugs is also needed because no effective licensed alternatives are available.⁷ Various studies in NICUs have estimated that 90% of neonatal patients receive at least one off-label prescription, and 60% of all prescriptions are off-label.⁸ In Spain, similar rates were recently reported by Arocas *et al.*⁹ Some studies have concluded that a high percentage of adverse drug reactions in paediatric populations is associated with off-label and/or unlicensed prescriptions.^{10–11}

The potential for severe harm resulting from medication errors in a NICU is high owing to use of the intravenous (IV) route, the common use of high-alert medications¹² and the often critical condition of neonates. Infusion-concentration standardisation reduces the number of drug preparation steps, simplifying and accelerating the process. In 2004, the US Joint Commission recommended that infusion practice should be standardised across paediatric care.¹³ In 2011, the Institute for Safe Medication Practices (ISMP) and the Vermont Oxford Network published a list of standard concentration recommendations for drug infusions in neonatal care.¹⁴ Nonetheless, there is still no general agreement on optimal concentrations, although several health organisations are making efforts to develop a consensus.

The main objective of this study was to describe the IV drugs most frequently used in Spanish NICUs, their preparation, and the implementation rate of standardised concentration infusions. Subsequent analysis of this first evaluation will be useful for designing strategies to improve neonatal patient safety in intravenous therapy.

MATERIALS AND METHODS

This was an observational, multicentre study based on a survey approved by the ethics committees of all participating hospitals. An invitation to participate in the survey was sent electronically through the Maternal and Child Health and Development Research Network (SAMID) to all members, and four NICUs consented to participate in the study. The other participating NICUs corresponded to units that had previously worked with this research group and had expressed their interest in continuing collaborating. According to the care-level classification published by the Spanish Neonatal Society,¹⁵ the nine participating NICUs were level III (that is, part of a referral hospital with paediatric and



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maternity services, handling at least 1000 deliveries a year and caring for complex, critically ill neonates, including those requiring mechanical ventilation) – two were level IIIA, four were level IIIB and three were level IIIC.

A survey on drugs administered intravenously in NICUs was designed by two hospital pharmacists, specialised in medication errors and patient safety, and checked by a neonatologist and a neonatal nurse specialist. The aim of the survey was to collect information related to drug preparation and administration phases. The survey covered 28 IV drugs frequently used in NICUs, obtained from the 2016 drug consumption records of one hospital, and allowed respondents to add any additional drugs they considered to be in frequent use in their units. The drugs of interest were those used for continuous and/or intermittent infusion. Data collected for each drug were: concentration, diluent, container, time of administration, and whether the mixture was prepared in the hospital pharmacy. The survey was conducted via email during January and February 2018. In the event of ambiguous or implausible responses, participants were contacted via email or phone for clarification. Hospitals that were geographically close to the main researcher were attended in person.

We performed a descriptive analysis of the IV drugs reported, of drug preparation, and of the use of standard concentration infusions. For each drug, we documented its Anatomical Therapeutic Chemical (ATC) classification, its licence status and whether it was a high-alert medication. The ATC codes were retrieved from the ATC index of the *WHO Collaborating Centre for Drug Statistics Methodology*. Licence status was analysed by consulting the formal product specifications approved by the Spanish Agency of Medicines and Health Products (AEMPS). Drugs were classified by their use-status as approved, off-label or unlicensed, according to the criteria published by Arocas *et al.*⁹ Drugs used outside the terms of the marketing authorisation were considered off-label use. Drugs without an AEMPS marketing authorization were defined as unlicensed use, including imported drugs and magistral formulations. Magistral formulation was defined as a compounded drug that was used only when commercially available drug products did not meet the needs of the patient; these were elaborated by modification of licenced drugs, or prepared from raw materials and chemicals used as drugs. *High-alert medications* have been defined by the ISMP as “drugs that bear a heightened risk of causing significant patient harm when they are used in error”, and for this study we considered the list published by the Institute in February 2018.¹²

Regarding standardisation, only infusion drugs with one or more standard concentrations established in the NICU were considered as standardised drugs. Continuous variables were reported using means and ranges.

RESULTS

Data were collected on a total of 69 different drugs, with a mean of 33 drugs (range 25–57) per NICU. Notably, 14 drugs were used in all NICUs (table 1), 30 drugs in more than 50% of units and 25 in only one.

Drugs were classified by the type of infusion for which they were used: continuous (n=26), intermittent (n=39) or both (n=4). In each hospital, more than 50% of the drugs used were administered by intermittent infusion (mean 61%; range 53%–69%). Considering their ATC codes, most of the drugs were in Group J (Anti-infectives for systemic use; n=27, 39%), followed by Group C (Cardiovascular system; n=20, 29%) and Group N (Nervous system; n=13, 19%). The most frequently-reported

Table 1 Drugs used in all neonatal intensive care units, their ATC codes, their high-alert medication status, and the type of infusion administered

Drugs	ATC code	High-alert medications	Continuous Infusion (CI) or Intermittent Infusion (II)
Acyclovir	J	No	II
Alprostadil	C	Yes	CI
Amphotericin B liposome	J	Yes	II
Dobutamine	C	Yes	CI
Dopamine	C	Yes	CI
Epinephrine	C	Yes	CI
Fentanyl	N	Yes	CI
Ganciclovir	J	No	II
Insulin	A	Yes	CI
Levetiracetam	N	No	II
Midazolam	N	Yes	CI
Paracetamol	N	No	II
Phenytoin	N	No	II
Vancomycin	J	No	II

ATC code: Group J (Anti-infectives for systemic use); Group C (Cardiovascular system); Group N (Nervous system) and Group A (Alimentary tract and metabolism). Continuous Infusion (CI) or Intermittent Infusion (II).

drugs in each case were: amphotericin B liposome, cefotaxime, ganciclovir, gentamicin and vancomycin from Group J; epinephrine, alprostadil, dobutamine, dopamine, furosemide, norepinephrine and milrinone from Group C; and fentanyl, midazolam, paracetamol and phenytoin from Group N.

One-third (n=23, 33%) of the drugs were used off-label. Less than two-thirds of the drugs were approved in neonates and these corresponded to two groups: those whose label information specified their indication in neonates (n=29, 42%) and those approved for paediatric use and not contraindicated in neonates (n=13, 19%). Further, one drug was cited by respondents (sodium nitroprusside) for which the safety and effectiveness are not well established in young children (according to its label). Finally, three drugs were cited that were classified as unlicensed in Spain: two being imported (flucytosine and clonidine) and one magistral formulation (caffeine anhydrous).

As many as 38% (n=26) of the drugs corresponded to high-alert medications, according to the ISMP list: adrenergic agonists and antagonists, antiarrhythmics, inotropic drugs, anaesthetics and sedation agents, neuromuscular-blocking medicines, opioids, heparin, amphotericin B liposome, insulin, sodium nitroprusside and alprostadil. Nearly all (96%) high-alert medications were administered as a continuous infusion.

Regarding drug concentration standardisation (figure 1), the mean proportion of standardised drug use was 63.5% (range 39.3–96.5%). Standard-concentration implementation rates were lower for continuous infusions, as they were standardised only in six NICUs. One hospital reported that all continuous infusions were standardised, and in three further hospitals more than 80% of continuous infusions were standardised. Regarding intermittent infusions, in two hospitals 100% had standard concentration implementation, and in the others the implementation was more than 50% (mean 74.1%; range 50%–100%).

Regarding diluents, dextrose 5% or sodium chloride 0.9% were used in almost all of the standardised concentration drugs. Other diluents were only used in one hospital, and these were dextrose 2.5% in sodium chloride 0.15% and dextrose 5% with

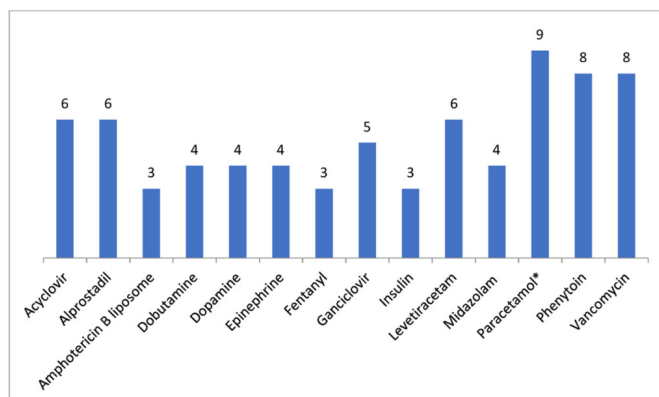


Figure 1 Number of NICUs with standardised concentration implemented of the fourteen IV drugs reported by all NICUs. *Commercial presentation.

albumin. In all hospitals, pump syringes were used as a container. In the case of photosensitive drugs, an opaque syringe was used.

Infusions were more commonly prepared on wards than in pharmacies. In one hospital, all continuous infusion medications were routinely prepared in the hospital pharmacy; in another, continuous infusions of five drugs (dopamine, dobutamine, midazolam, alprostadil and fentanyl) were routinely handled in the pharmacy service if they were individualised, whereas standard concentration mixtures were compounded in the NICU. In both of these hospitals, however, preparation instructions were available for nurses on wards to use when a continuous infusion treatment was urgent or started outside the pharmacy opening hours. In relation to intermittent infusions, three drugs were reported as being prepared in the hospital pharmacy in some hospitals: acyclovir in two, ganciclovir in seven, and amphotericin B liposome in seven.

DISCUSSION

This is the first descriptive study based on a survey of IV drug use in Spanish NICUs. Our survey focused on intermittent and continuous infusion preparation and administration practices, as activities related to these two areas are associated with a particularly high risk of patient safety events.

In our study, a total of 69 different drugs were reported as being frequently used in NICUs. The list of drugs most commonly used, according to our results, is consistent with data previously published in the literature, with the following ATC groups being dominant: anti-infectives for systemic use, cardiovascular system preparations, and drugs for the nervous system.^{7-9 16} Despite the reduced number of samples used in the present study, a substantial difference in terms of the IV drugs reported would not be expected across other Spanish NICUs since high healthcare level NICUs (as presented in this study) cover a consistent set of drugs. Discrepancies between NICUs in term of the drugs they frequently used were identified mainly between different healthcare levels, since drugs such as dexmedetomidine, ketamine and cisatracurium were only reported by NICUs of level IIIC. From ATC Groups C and N, 75% and 46.1% of drugs, respectively, corresponded to high-alert medications on the ISMP list.¹² Notably, 26 drugs were classified as high-alert, indicating a considerable risk of potential harm during their administration.

Regarding drug licence status, a lack of specific drugs and labelling recommendations for neonates is a long-standing problem worldwide, which may lead to adverse drug reactions^{7 10} and medication errors.^{11 17} A systematic review indicated that

newborn infants received the highest proportions of off-label and unlicensed drugs.¹⁸ The main reasons cited for off-label prescribing were lack of information on paediatric use and the use of drugs outside the permitted age range or the established dose range, whilst the main factor associated with prescribing unlicensed drugs was the modification of licensed drugs, and in this sense the local tradition of producing compounded preparations constitutes an important factor.

In our study, we collected data on IV drugs frequently used in NICUs using a survey, so we obtained the number of drugs used off-label and not prescription rates. Consequently, our results are difficult to compare with those of other published studies. Nevertheless, our findings support the view that there is a lack of controlled clinical trials concerning neonatal licencing of drugs widely used in very young infants, in that a third of drugs commonly used in the NICUs were not approved in neonates. The lack of approval for neonatal use responds to the low incentive for pharmaceutical companies to develop drugs and dosing guidelines for the neonatal population, but does not imply that a medicine is contraindicated or disapproved. In Europe, a priority list of drugs requiring data in the paediatric age group was established by the European Medicines Agency and is updated every year.

The profile of drug use identified in the study reflects the complexity of treatments in NICUs, and underlines both the high potential for severe harm if medication errors occur and the need to prioritise the implementation of safety strategies in this setting. Taking the ECLIPSE study in the UK as a reference,¹⁹ we explored and documented the variety of IV infusion practices in our hospitals as a first step in order to deliver recommendations for best practice in different situations. Interventions for preventing errors in the drug-handling process include: barcode-assisted medication administration systems, smart infusion pumps, color-coded medication safety systems, distraction-free spaces for handling, training, and standardisation of drug concentrations.^{4 20} We have focused on analysing the rate of implementation of this last measure in Spanish NICUs.

Larsen *et al*²¹ found a 73% reduction in reported errors after the implementation of 32 standard drug concentrations (together with the use of smart syringe pumps and “human-engineered” medication labels). Further, Hilmas *et al*²² described the elimination of errors related to exceeding the maximum drug concentration after the implementation of a ‘computerised prescriber order entry system’ that used standardised concentrations for continuous infusions. In our study, all NICUs used standard drug concentrations in at least 39% of the medications frequently administered in their wards. In the UK, results from a national survey on standard IV infusion concentration²³ was the basis of a multi-professional collaborative initiative, working towards a national consensus on standard concentration infusions in paediatric and neonatal care. That study found that 40% of paediatric and neonatal units had established standard concentrations for infusion drugs, being more common in units caring for neonatal patients (61%). A total of 46 drugs were standardised, with 143 variations (considering only concentration). In 2017, this group published a consensus on fixed concentrations for the 20 most commonly infused medicines, after a Delphi study across the country.²⁴ However, implementation rates in European hospitals remain far below those in the USA, where (for the last decade) accreditation has been linked to implementation and institutional initiatives have been promoted (such as *Standardise 4 Safety*, the first national, interprofessional effort to standardise oral and IV medication concentrations).

Regarding the type of infusion, intermittent infusion drugs were standardised to some degree in all NICUs, with a mean of 74.1%

being standardised, whereas continuous infusion medicines were standardised in only six NICUs. According to the ISMP list, 84% of continuous infusion drugs were on the high-alert list, and standardisation of those drugs should be prioritised.

As drug preparation is a high-risk process in NICUs, we collected data on how often medicines were prepared in the pharmacy. Intermittent infusion drugs, which (according to safety and economic criteria) should be handled in a hospital pharmacy, were indeed prepared there; an example is ganciclovir, which is classified as a hazardous drug for health workers due to its potential mutagenic and teratogenic toxicity.²⁵ In contrast, continuous infusions were prepared in the pharmacy service in just two hospitals.

Medicine preparation in a hospital pharmacy reduces medication errors and the risk of microbiological contamination, as it ensures that processes are carried out to high safety standards (with dedicated clean-working areas, adequate equipment, specific training and minimal distractions, among other factors). Campino *et al*²⁶ measured IV preparation error rates and compared those in NICUs with those in the hospital pharmacy. In samples from the hospital pharmacy, no calculation errors were detected but accuracy errors were found in 38.3% of samples; it was concluded that calculation errors could be avoided with standard concentration protocols but that accuracy depends on good practices in the preparation process. In a later study, after the implementation of a standard-concentration protocol in all NICUs and educational interventions with nurses, calculation errors were eliminated and accuracy errors decreased in both settings.²⁷ Samples did not undergo microbiological testing in any of these studies.

In Spain, the Ministry of Health published recommendations setting out preparation requirements, helping professionals with the decision of where to prepare medications: clinical areas vs hospital pharmacy clean rooms.²⁸ In those recommendations, the need for calculations and a greater number of steps for drug preparation are among the criteria indicating that a drug should be prepared in the pharmacy. Nevertheless, our study reveals a low rate of pharmacy handling of neonatal infusions, attributable to the heavy workload in pharmacies and the lack of resources (normally implying limited opening hours). As some studies have shown,^{27–29} educational strategies are an effective measure for preventing preparation errors and require fewer resources than central pharmacy preparation and, hence, training could be a suitable alternative.

The main limitation of our study is the risk of bias inherent to the method of data collection, since the quality or veracity of responses to the survey cannot be assessed and the data collected on the survey were limited. Despite our sample being representative of the three highest neonatal care levels (according to the Spanish Neonatal Society¹⁵), meaning that variation in the IV drugs reported would not be expected, a wider variation of standard concentrations and administration practices might be found if a greater number of participating NICUs were surveyed. This would be necessary for a full picture of the Spanish situation.

CONCLUSION

Our study confirms the high-risk profile of IV therapy used in neonates and the potential risk of harm if medication errors occur. Despite the ISMP in Spain recommending concentration standardisation to prevent medication errors in children, this practice is not very common in our NICUs, particularly in the case of continuous infusions. Further data would be necessary to design a suitable intervention in our country, but institutional initiatives are needed to promote the implementation of this recommendation.

What this paper adds

What is already known on this subject

- ▶ Intravenous drug use in newborns is frequent and prone to medication errors.
- ▶ Error-reduction strategies, such as concentration standardisation and pharmacy preparation, are recommended for infusion drugs.

What this study adds

- ▶ We obtained a partial picture of intravenous drug use in Spanish NICUs, and the high-risk profile of intravenous drug use in neonates was confirmed.
- ▶ Standardised medication concentration was implemented to some degree in all participating NICUs but additional efforts are required, especially for continuous infusion.

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Contributors All authors made substantial contributions to conception and design of the study, and/or acquisition of data, and/or analysis and interpretation of data; all authors participated in drafting or critically revising the manuscript for important intellectual content; all authors gave final approval of the version submitted; and all authors agree to be accountable for all aspects of the work in ensuring that questions related to the accuracy or integrity of any part of the work are appropriately investigated and resolved. AdB and AC: Conceived the idea, developed the project, designed and disseminated the survey, drafted the initial manuscript and revisions and approved the final manuscript to be submitted. Both agree to be accountable for all aspects of the work in ensuring that questions related to the accuracy or integrity of any part of the work are appropriately investigated and resolved. AF and SM: Collaborated in the data collection, analysis and interpretation of data. Drafted the initial manuscript and revisions and approved the final manuscript to be submitted. Agrees to be accountable for all aspects of the work in ensuring that questions related to the accuracy or integrity of any part of the work are appropriately investigated and resolved. MdM, EG and BL: Revised and approved the final manuscript. Agrees to be accountable for all aspects of the work in ensuring that questions related to the accuracy or integrity of any part of the work are appropriately investigated and resolved.

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